

Derivation of Einstein Field Equations

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July 27, 2026

1 Problem statement:

Derive the field equations in vacuum from the Einstein-Hilbert action.

- (a) Prove the variation of the square root of the determinant of the metric is

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$$

- (b) Prove the variation of the Riemann tensor is

$$\delta R^\rho{}_{\mu\lambda\nu} = \nabla_\lambda(\delta\Gamma^\rho_{\nu\mu}) - \nabla_\nu(\delta\Gamma^\rho_{\lambda\mu})$$

- (c) Prove the variation of the Christoffel symbol is

$$\delta\Gamma^\sigma_{\mu\nu} = -\frac{1}{2}\left[2g_{\lambda(\mu}\nabla_{\nu)}(\delta g^{\lambda\sigma}) - g_{\mu\alpha}g_{\nu\beta}(\nabla^\sigma\delta g^{\alpha\beta})\right]$$

- (d) With parts (b) and (c), prove that the total variation of the Ricci tensor vanishes:

$$\delta S = \int \sqrt{-g} d^4x g^{\mu\nu} \delta R_{\mu\nu} = \int \sqrt{-g} d^4x g^{\mu\nu} \delta R^\rho{}_{\mu\rho\nu} = 0$$

- (e) Finally, put all of these steps together to derive Einstein's equation in vacuum:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0$$

Before you proceed, try to do this yourself! It's a good exercise. Check your work against the solution on the following pages.

2 Solution:

(a) Prove the variation of the square root of the determinant of the metric is

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$$

Proof. For any square matrix M with non-vanishing determinant,

$$\begin{aligned}\ln(\det M) &= \text{Tr}(\ln M) \\ \Rightarrow \frac{1}{\det M}\delta(\det M) &= \text{Tr}(M^{-1}\delta M)\end{aligned}$$

Taking the matrix M to be the metric $g_{\mu\nu}$ we have

$$\frac{1}{g}\delta g = \text{Tr}(g^{\mu\nu}\delta g_{\mu\nu}) = g^{\mu\nu}\delta g_{\mu\nu}$$

Now since the variation of the Kronecker delta is zero,

$$0 = \delta(\delta^\nu_\nu) = \delta(g_{\mu\nu}g^{\mu\nu}) = (\delta g_{\mu\nu})g^{\mu\nu} + g_{\mu\nu}(\delta g^{\mu\nu}) \Rightarrow g^{\mu\nu}\delta g_{\mu\nu} = -g_{\mu\nu}\delta g^{\mu\nu}$$

So we can rewrite the variation of the metric determinant as

$$\delta g = g(g^{\mu\nu}\delta g_{\mu\nu}) = -g(g_{\mu\nu}\delta g^{\mu\nu})$$

Therefore

$$\begin{aligned}\delta\sqrt{-g} &= -\frac{1}{2}\frac{1}{\sqrt{-g}}\delta g \\ &= \frac{1}{2\sqrt{-g}}g g_{\mu\nu}\delta g^{\mu\nu} \\ &= \frac{1}{2}\frac{(-\sqrt{-g}\sqrt{-g})}{\sqrt{-g}}g_{\mu\nu}\delta g^{\mu\nu} \\ \boxed{\delta\sqrt{-g} &= -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}}\end{aligned}$$

□

(b) Prove the variation of the Riemann tensor is

$$\delta R^\rho_{\mu\lambda\nu} = \nabla_\lambda (\delta\Gamma^\rho_{\nu\mu}) - \nabla_\nu (\delta\Gamma^\rho_{\lambda\mu})$$

Proof. Consider variations of the connection $\Gamma^\rho_{\nu\mu} \rightarrow \Gamma^\rho_{\nu\mu} + \delta\Gamma^\rho_{\nu\mu}$. Since $\delta\Gamma^\rho_{\nu\mu}$ is the difference between two connections, it is a tensor and we can take a covariant derivative of it. If we take the covariant derivative of this variation,

$$\nabla_\lambda (\delta\Gamma^\rho_{\nu\mu}) = \partial_\lambda (\delta\Gamma^\rho_{\nu\mu}) + \Gamma^\rho_{\lambda\sigma}\delta\Gamma^\sigma_{\nu\mu} - \Gamma^\sigma_{\lambda\nu}\delta\Gamma^\rho_{\sigma\mu} - \Gamma^\sigma_{\lambda\mu}\delta\Gamma^\rho_{\nu\sigma}$$

while

$$\nabla_\nu (\delta\Gamma_{\lambda\mu}^\rho) = \partial_\nu (\delta\Gamma_{\lambda\mu}^\rho) + \Gamma_{\nu\sigma}^\rho \delta\Gamma_{\lambda\mu}^\sigma - \Gamma_{\nu\lambda}^\sigma \delta\Gamma_{\sigma\mu}^\rho - \Gamma_{\nu\mu}^\sigma \delta\Gamma_{\lambda\sigma}^\rho$$

Then, putting these two expressions together, we find that

$$\begin{aligned} \nabla_\lambda (\delta\Gamma_{\nu\mu}^\rho) - \nabla_\nu (\delta\Gamma_{\lambda\mu}^\rho) &= \partial_\lambda (\delta\Gamma_{\nu\mu}^\rho) + \Gamma_{\lambda\sigma}^\rho \delta\Gamma_{\nu\mu}^\sigma - \Gamma_{\lambda\nu}^\sigma \delta\Gamma_{\sigma\mu}^\rho - \Gamma_{\lambda\mu}^\sigma \delta\Gamma_{\nu\sigma}^\rho \\ &\quad - \left(\partial_\nu (\delta\Gamma_{\lambda\mu}^\rho) + \Gamma_{\nu\sigma}^\rho \delta\Gamma_{\lambda\mu}^\sigma - \Gamma_{\nu\lambda}^\sigma \delta\Gamma_{\sigma\mu}^\rho - \Gamma_{\nu\mu}^\sigma \delta\Gamma_{\lambda\sigma}^\rho \right) \\ &= \partial_\lambda (\delta\Gamma_{\nu\mu}^\rho) - \partial_\nu (\delta\Gamma_{\lambda\mu}^\rho) + \Gamma_{\lambda\sigma}^\rho \delta\Gamma_{\nu\mu}^\sigma - \Gamma_{\lambda\mu}^\sigma \delta\Gamma_{\nu\sigma}^\rho \\ &\quad - (\Gamma_{\nu\sigma}^\rho \delta\Gamma_{\lambda\mu}^\sigma - \Gamma_{\nu\mu}^\sigma \delta\Gamma_{\lambda\sigma}^\rho) \\ &= \partial_\lambda (\delta\Gamma_{\nu\mu}^\rho) - \partial_\nu (\delta\Gamma_{\lambda\mu}^\rho) + \Gamma_{\lambda\sigma}^\rho \delta\Gamma_{\nu\mu}^\sigma + \Gamma_{\nu\mu}^\sigma \delta\Gamma_{\lambda\sigma}^\rho \\ &\quad - \Gamma_{\lambda\mu}^\sigma \delta\Gamma_{\nu\sigma}^\rho - \Gamma_{\nu\sigma}^\rho \delta\Gamma_{\lambda\mu}^\sigma \\ &= \partial_\lambda (\delta\Gamma_{\nu\mu}^\rho) - \partial_\nu (\delta\Gamma_{\lambda\mu}^\rho) + \delta (\Gamma_{\lambda\sigma}^\rho \Gamma_{\nu\mu}^\sigma) - \delta (\Gamma_{\lambda\mu}^\sigma \Gamma_{\nu\sigma}^\rho) \\ &= \delta (\partial_\lambda \Gamma_{\nu\mu}^\rho - \partial_\nu \Gamma_{\lambda\mu}^\rho + \Gamma_{\lambda\sigma}^\rho \Gamma_{\nu\mu}^\sigma - \Gamma_{\lambda\mu}^\sigma \Gamma_{\nu\sigma}^\rho) \end{aligned}$$

$$\nabla_\lambda (\delta\Gamma_{\nu\mu}^\rho) - \nabla_\nu (\delta\Gamma_{\lambda\mu}^\rho) = \delta R_{\mu\lambda\nu}^\rho$$

□

(c) Prove the variation of the Christoffel symbol is

$$\delta\Gamma_{\mu\nu}^\sigma = -\frac{1}{2} \left[2g_{\lambda(\mu} \nabla_{\nu)} (\delta g^{\lambda\sigma}) - g_{\mu\alpha} g_{\nu\beta} (\nabla^\sigma \delta g^{\alpha\beta}) \right]$$

Proof. It will be useful in what follows to know that by equation 4.56 of Carroll, $\delta g_{\mu\nu} = -g_{\mu\alpha} g_{\nu\beta} \delta g^{\alpha\beta}$ and hence $g^{\sigma\lambda} \delta g_{\nu\lambda} = -g_{\nu\lambda} \delta g^{\sigma\lambda}$. By definition, the Christoffel symbol is given by

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\lambda} (\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})$$

So the variation of the Christoffel symbol is given by

$$\begin{aligned}
\delta\Gamma_{\mu\nu}^{\sigma} &= \frac{1}{2}g^{\sigma\lambda}(\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu})) + \frac{1}{2}(\delta g^{\sigma\lambda})(\partial_{\mu}g_{\nu\lambda} + \partial_{\nu}g_{\lambda\mu} - \partial_{\lambda}g_{\mu\nu}) \\
&= \frac{1}{2}g^{\sigma\lambda}(\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu})) + (\delta g^{\sigma\lambda})\frac{1}{2}g_{\lambda\rho}g^{\rho\alpha}(\partial_{\mu}g_{\nu\alpha} + \partial_{\nu}g_{\alpha\mu} - \partial_{\alpha}g_{\mu\nu}) \\
&= \frac{1}{2}g^{\sigma\lambda}(\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu})) + (\delta g^{\sigma\lambda})g_{\lambda\rho}\Gamma_{\mu\nu}^{\rho} \\
&= \frac{1}{2}g^{\sigma\lambda}(\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu})) - \frac{1}{2}2(\delta g_{\lambda\rho})g^{\sigma\lambda}\Gamma_{\mu\nu}^{\rho} \\
&= \frac{1}{2}g^{\sigma\lambda}[\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu}) - 2\Gamma_{\mu\nu}^{\rho}(\delta g_{\lambda\rho})] \\
&= \frac{1}{2}g^{\sigma\lambda}[\partial_{\mu}(\delta g_{\nu\lambda}) + \partial_{\nu}(\delta g_{\lambda\mu}) - \partial_{\lambda}(\delta g_{\mu\nu}) \\
&\quad - \Gamma_{\mu\nu}^{\rho}\delta g_{\lambda\rho} - \Gamma_{\mu\nu}^{\rho}\delta g_{\lambda\rho} + \Gamma_{\lambda\mu}^{\rho}\delta g_{\rho\nu} - \Gamma_{\lambda\mu}^{\rho}\delta g_{\rho\nu} + \Gamma_{\lambda\nu}^{\rho}\delta g_{\rho\mu} - \Gamma_{\lambda\nu}^{\rho}\delta g_{\rho\mu}] \\
&= \frac{1}{2}g^{\sigma\lambda}\left[\partial_{\mu}(\delta g_{\nu\lambda}) - \Gamma_{\lambda\mu}^{\rho}\delta g_{\rho\nu} - \Gamma_{\mu\nu}^{\rho}\delta g_{\lambda\rho} + \partial_{\nu}(\delta g_{\lambda\mu}) - \Gamma_{\lambda\nu}^{\rho}\delta g_{\rho\mu} - \Gamma_{\mu\nu}^{\rho}\delta g_{\lambda\rho} \right. \\
&\quad \left. - \partial_{\lambda}(\delta g_{\mu\nu}) + \Gamma_{\lambda\mu}^{\rho}\delta g_{\rho\nu} + \Gamma_{\lambda\nu}^{\rho}\delta g_{\rho\mu}\right] \\
&= \frac{1}{2}g^{\sigma\lambda}[\nabla_{\mu}(\delta g_{\nu\lambda}) + \nabla_{\nu}(\delta g_{\mu\lambda}) - \nabla_{\lambda}(\delta g_{\mu\nu})] \\
&= -\frac{1}{2}\left[-g^{\sigma\lambda}\nabla_{\mu}(\delta g_{\nu\lambda}) - g^{\sigma\lambda}\nabla_{\nu}(\delta g_{\mu\lambda}) + g^{\sigma\lambda}\nabla_{\lambda}(\delta g_{\mu\nu})\right] \\
&= -\frac{1}{2}\left[-\nabla_{\mu}(g^{\sigma\lambda}\delta g_{\nu\lambda}) - \nabla_{\nu}(g^{\sigma\lambda}\delta g_{\mu\lambda}) + \nabla^{\sigma}(\delta g_{\mu\nu})\right] \\
&= -\frac{1}{2}\left[-\nabla_{\mu}(-g_{\nu\lambda}\delta g^{\sigma\lambda}) - \nabla_{\nu}(-g_{\mu\lambda}\delta g^{\sigma\lambda}) + \nabla^{\sigma}(\delta g_{\mu\nu})\right] \\
&= -\frac{1}{2}\left[g_{\nu\lambda}\nabla_{\mu}(\delta g^{\sigma\lambda}) + g_{\mu\lambda}\nabla_{\nu}(\delta g^{\sigma\lambda}) + \nabla^{\sigma}(-g_{\mu\alpha}g_{\nu\beta}\delta g^{\alpha\beta})\right] \\
&\boxed{\delta\Gamma_{\mu\nu}^{\sigma} = -\frac{1}{2}\left[2g_{\lambda(\nu}\nabla_{\mu)}(\delta g^{\sigma\lambda}) - g_{\mu\alpha}g_{\nu\beta}\nabla^{\sigma}(\delta g^{\alpha\beta})\right]}
\end{aligned}$$

□

(d) With parts (b) and (c), prove that the total variation of the Ricci tensor vanishes:

$$\delta S = \int \sqrt{-g} d^4x g^{\mu\nu} \delta R_{\mu\nu} = \int \sqrt{-g} d^4x g^{\mu\nu} \delta R^{\rho}_{\mu\rho\nu} = 0$$

Proof. The Ricci tensor $R_{\mu\nu} = R^{\rho}_{\mu\rho\nu} \Rightarrow \delta R_{\mu\nu} = \delta R^{\rho}_{\mu\rho\nu}$. Then by part (b), the variation of this Riemann tensor is

$$\delta R^{\rho}_{\mu\rho\nu} = \nabla_{\rho}(\delta\Gamma^{\rho}_{\nu\mu}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\rho\mu})$$

Applying $g^{\mu\nu}$ onto that:

$$\begin{aligned} g^{\mu\nu} \delta R^\rho_{\mu\rho\nu} &= g^{\mu\nu} \nabla_\rho (\delta \Gamma^\rho_{\nu\mu}) - g^{\mu\nu} \nabla_\nu (\delta \Gamma^\rho_{\rho\mu}) \\ &= g^{\mu\nu} \nabla_\sigma (\delta \Gamma^\sigma_{\nu\mu}) - g^{\mu\sigma} \nabla_\sigma (\delta \Gamma^\rho_{\rho\mu}) \\ g^{\mu\nu} \delta R_{\mu\nu} &= \nabla_\sigma [g^{\mu\nu} (\delta \Gamma^\sigma_{\nu\mu}) - g^{\mu\sigma} (\delta \Gamma^\rho_{\rho\mu})] \end{aligned}$$

We can plug in an expression for the variation of each of the Christoffel symbols using part (c):

$$\begin{aligned} g^{\mu\nu} \delta R^\rho_{\mu\rho\nu} &= \nabla_\sigma [g^{\mu\nu} (\delta \Gamma^\sigma_{\nu\mu}) - g^{\mu\sigma} (\delta \Gamma^\rho_{\rho\mu})] \\ &= \nabla_\sigma \left[g^{\mu\nu} \left(-\frac{1}{2}(2)g_{\lambda(\nu} \nabla_{\mu)} \delta g^{\sigma\lambda} + \frac{1}{2}g_{\mu\alpha} g_{\nu\beta} \nabla^\sigma \delta g^{\alpha\beta} \right) \right. \\ &\quad \left. - g^{\mu\sigma} \left(-\frac{1}{2}(2)g_{\lambda(\rho} \nabla_{\mu)} \delta g^{\rho\lambda} + \frac{1}{2}g_{\rho\alpha} g_{\mu\beta} \nabla^\rho \delta g^{\alpha\beta} \right) \right] \\ &= \nabla_\sigma \left[-\frac{1}{2}(2)g^{\mu\nu} g_{\lambda(\nu} \nabla_{\mu)} \delta g^{\sigma\lambda} + \frac{1}{2}g^{\mu\nu} g_{\mu\alpha} g_{\nu\beta} \nabla^\sigma \delta g^{\alpha\beta} \right. \\ &\quad \left. + \frac{1}{2}(2)g^{\mu\sigma} g_{\lambda(\rho} \nabla_{\mu)} \delta g^{\rho\lambda} - \frac{1}{2}g^{\mu\sigma} g_{\rho\alpha} g_{\mu\beta} \nabla^\rho \delta g^{\alpha\beta} \right] \\ &= \nabla_\sigma \left[-\frac{1}{2}(2)g^{\mu\nu} g_{\lambda(\nu} \nabla_{\mu)} \delta g^{\sigma\lambda} + \frac{1}{2}\delta^\nu_\alpha g_{\nu\beta} \nabla^\sigma \delta g^{\alpha\beta} \right. \\ &\quad \left. + \frac{1}{2}(2)g^{\mu\sigma} g_{\lambda(\rho} \nabla_{\mu)} \delta g^{\rho\lambda} - \frac{1}{2}\delta^\sigma_\beta g_{\rho\alpha} \nabla^\rho \delta g^{\alpha\beta} \right] \\ &= \nabla_\sigma \left[-\frac{1}{2}(2)g^{\mu\nu} g_{\lambda(\nu} \nabla_{\mu)} \delta g^{\sigma\lambda} + \frac{1}{2}g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} \right. \\ &\quad \left. + \frac{1}{2}(2)g^{\mu\sigma} g_{\lambda(\rho} \nabla_{\mu)} \delta g^{\rho\lambda} - \frac{1}{2}\delta^\sigma_\beta \nabla_\alpha \delta g^{\alpha\beta} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-g^{\mu\nu} g_{\lambda\nu} \nabla_\mu \delta g^{\sigma\lambda} - g^{\mu\nu} g_{\lambda\mu} \nabla_\nu \delta g^{\sigma\lambda} + g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} \right. \\ &\quad \left. + (2)g_{\lambda(\rho} \nabla^{\sigma)} \delta g^{\rho\lambda} - \delta^\sigma_\beta \nabla_\alpha \delta g^{\alpha\beta} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-\delta^\mu_\lambda \nabla_\mu \delta g^{\sigma\lambda} - \delta^\nu_\lambda \nabla_\nu \delta g^{\sigma\lambda} + g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} + (2)g_{\lambda(\rho} \nabla^{\sigma)} \delta g^{\rho\lambda} - \delta^\sigma_\beta \nabla_\alpha \delta g^{\alpha\beta} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-2\nabla_\lambda \delta g^{\sigma\lambda} + g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} + g_{\lambda\rho} \nabla^\sigma \delta g^{\rho\lambda} + g_\lambda{}^\sigma \nabla_\rho \delta g^{\rho\lambda} - \delta^\sigma_\beta \nabla_\alpha \delta g^{\alpha\beta} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-2\nabla_\lambda \delta g^{\sigma\lambda} + g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} + g_{\lambda\rho} \nabla^\sigma \delta g^{\rho\lambda} + \nabla_\rho \delta g^{\rho\sigma} - \nabla_\alpha \delta g^{\alpha\sigma} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-2\nabla_\lambda \delta g^{\sigma\lambda} + g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} + g_{\alpha\beta} \nabla^\sigma \delta g^{\beta\alpha} + \nabla_\alpha \delta g^{\alpha\sigma} - \nabla_\alpha \delta g^{\alpha\sigma} \right] \\ &= \frac{1}{2} \nabla_\sigma \left[-2\nabla_\lambda \delta g^{\sigma\lambda} + 2g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} \right] \\ g^{\mu\nu} \delta R_{\mu\nu} &= \nabla_\sigma \left[g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} - \nabla_\lambda \delta g^{\sigma\lambda} \right] \end{aligned}$$

Plugging this in to the integral we have

$$\begin{aligned}
\delta S &= \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} \\
&= \int d^4x \sqrt{-g} \nabla_\sigma \left[g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} - \nabla_\lambda \delta g^{\sigma\lambda} \right] \\
\delta S &= \int_{\partial M} d^3y \sqrt{-\gamma} n_\sigma \left[g_{\alpha\beta} \nabla^\sigma \delta g^{\alpha\beta} - \nabla_\lambda \delta g^{\sigma\lambda} \right]
\end{aligned}$$

by Stokes' Theorem (E.14 of Carroll). Here ∂M is the boundary of the manifold, γ is the induced metric, and n^σ is the unit normal to the boundary. Because this was an integral with respect to the volume element of the covariant divergence of a vector, our boundary condition is at infinity. We can set the variation of the metric to be zero at infinity, and then this term vanishes.

$$\Rightarrow \boxed{\delta S = \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} = 0}$$

□

(e) Finally, put all of these steps together to derive Einstein's equation.

Consider the Einstein-Hilbert action given in class,

$$S_H = \int d^4x \sqrt{-g} R = \int d^4x \sqrt{-g} g^{\mu\nu} R_{\mu\nu}$$

If we vary this action with respect to the metric,

$$\delta S_H = \int d^4x \left[g^{\mu\nu} R_{\mu\nu} \delta \sqrt{-g} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} \right]$$

By part (a) of this problem, we know that the first term can be rewritten as

$$g^{\mu\nu} R_{\mu\nu} \delta \sqrt{-g} = R \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} R \delta g^{\mu\nu}$$

By part (d) of this problem, we know that the third term vanishes,

$$\int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} = 0$$

Therefore,

$$\begin{aligned}
\delta S_H &= \int d^4x \left[g^{\mu\nu} R_{\mu\nu} \delta \sqrt{-g} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} \right] \\
&= \int d^4x \left[-\frac{1}{2} \sqrt{-g} g_{\mu\nu} R \delta g^{\mu\nu} + \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} \right] \\
\delta S_H &= \int d^4x \sqrt{-g} \left[R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right] \delta g^{\mu\nu}
\end{aligned}$$

Stationary points are those for which

$$\frac{\delta S_H}{\delta g^{\mu\nu}} = \sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0$$

and since the determinant of the metric is nonzero, we can equivalently write this as

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g^{\mu\nu}} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0$$

In vacuum, the stress-energy tensor is simply $T_{\mu\nu} = 0$, so we have recovered the Einstein Equations in vacuum:

$$\boxed{R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0}$$